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Frame assembly for windows and sliding doors

Abstract

A frame assembly, for use in window and door applications configured to retain a panel, comprising a frame with a cover assembly and a support assembly having a track assembly, and a sill assembly and a header assembly both interconnected to a securing surface. The cover assembly is interconnected to the support assembly. The sill assembly comprises a roller member disposed in supporting relation to the track assembly, and the header assembly is disposed in supporting relation to the support assembly so that the frame is movable relative to both the header assembly and the sill assembly. At least a portion of the frame comprises a predetermined thickness and a predetermined width. The predetermined thickness is substantially less than the predetermined width so that both cooperatively and concurrently enhance viewing through the panel and the frame's stability.

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Background/Summary

CLAIM OF PRIORITY (1) The present application is a Continuation to a currently pending Non-Provisional patent application having Ser. No. 17/845,899, filed Jun. 21, 2022 which is set to mature into U.S. Pat. No. 12,012,799 on Jun. 18, 2024, which is a Continuation of U.S. patent application Ser. No. 17/158,608 filed Jan. 26, 2021, which matured into U.S. Pat. No. 11,365,582 on Jun. 21, 2022, which is a Continuation-in-Part of U.S. patent application Ser. No. 16/654,871 filed Oct. 16, 2019, which matured into U.S. Pat. No. 10,900,273 on Jan. 26, 2021, and is a Continuation-in-Part to an abandoned Non-Provisional patent application having Ser. No. 15/648,666 and a filing date of Jul. 13, 2017 the contents of which are incorporated herein by reference in their entireties.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention is directed to a frame assembly comprising a frame configured to retain a panel for use in windows and doors. The frame of the present invention comprises a predetermined thickness having a reduced dimension relative to an increased dimension of a predetermined width of the frame. The predetermined thickness and width of the frame cooperatively enhance viewing through the panel and stability of the frame. The frame assembly of the present invention may also be configured for use in sliding glass doors having multiple tracks.

Description of the Related Art

(2) One important aesthetic feature of a window or a sliding glass door is the ability to minimize the visible area of the supporting frame and to maximize the visible area of the panel with the aim of enhancing viewing through the panel. Additionally, structural integrity is a necessary characteristic of a window or door frame assembly in most types of construction applications. More specifically, frame assemblies with impact resistant properties are necessary in many circumstances. There exists in the industry window and door frame assemblies that are impact resistant. Nevertheless, some of the problems with existing impact resistant assemblies are that they often will have a thicker size frame in order to achieve impact resistance. Consequently, portions of the thicker size frame may overlap with a window or door panel to an extent, at least partially obstructing viewing through the panel and reducing visibility. Other problems of some existing frame assemblies are that they often have a thinner size frame that may compromise structural stability or impact resistance.

(3) Thus, there is a need to provide a frame assembly for use in doors and windows having a frame of a size that enhances viewing and at the same time maintains structural integrity and impact resistant properties.

SUMMARY OF THE INVENTION

(4) The present invention is directed to a frame assembly for use in a variety of window and door applications. The frame assembly of the present invention is intended for use in windows and sliding glass doors of buildings, but may be used in other settings. The frame is generally structured to retain a panel, which may be at least partially formed of transparent or translucent material. In preferred embodiments according to the present invention, the frame enhances or at least significantly reduces obstructions to viewing through the panel while at the same time maintaining the stability of the frame. Generally, at least a portion of the frame, and in some embodiments a majority or substantially all of the frame, comprises a predetermined thickness and a predetermined width wherein the predetermined thickness has a reduced dimension relative to the predetermined width. In preferred embodiments of the frame assembly for use as a sliding glass door, the vertical portions of the frame have this predetermined configuration where the predetermined thickness is substantially less that the predetermined width. The predetermined thickness, being substantially

less than that the predetermined width, enhances viewing through the panel by minimizing the thickness of the frame portions which would normally obstruct a visible area of the frame. Accordingly, the visible area of the panel will be increased or maximized. In cooperation therewith, the relatively larger dimension of the predetermined width compensates for the relatively decreased predetermined thickness, enhancing or maintaining the stability of the frame.

(5) More specifically, the frame assembly of the present invention includes a frame generally having a support assembly and a cover assembly. The frame may also be in the form of a single piece. The support assembly is generally attached to a securing surface. The securing surface may be a wall, floor, soffit, ceiling, or other structure that supports the frame. The cover assembly may comprise a cover structure having a plurality of latching members and a retaining member, while the support assembly may have a plurality of corresponding latching structures. Together, the plurality of latching members and latching structures may be cooperatively configured to form a mating engagement to collectively interconnect the cover assembly and the support assembly. In preferred embodiments according to the present invention, the support assembly and the cover assembly, are cooperatively structured to retain a panel having varying dimensions. The cover assembly may comprise a retaining member. A cover clamp may be formed on an end of the retaining member. The retaining member or the cover clamp may be disposed in spaced relation with the cap segment of the support assembly. Both the retaining member and the cap segment, or the cover clamp and the cap segment, may be cooperatively configured to retain a panel having varying dimensions. Furthermore, a sealant member may be disposed between the cover clamp and the panel. The sealant member may be made of a variety of products including, but not limited to, elastomeric materials such as polyurethane caulking.

(6) Some embodiments according to the present invention may comprise a plurality of frames disposed in adjacent relation to one another. The frames may be interconnected to each other directly, or may comprise a strengthening member disposed between the frames. The strengthening member may be either unreinforced or internally reinforced. Additionally, the strengthening member may be attached to a securing surface as mentioned above. Other embodiments contemplate the use of an interconnecting member that may attach two adjacent frames, or that may attach a frame to a securing surface.

(7) The frame assembly according to the present invention may comprise a frame having a predetermined thickness of no more than substantially 1 and ½ inches. In some embodiments, the predetermined thickness of the frame may generally be about 1 to about 1 and ½ inches. In other preferred embodiments the predetermined thickness of the frame is about 1 and 3/16 inches. Conversely, the predetermined width of the frame may generally be about 2 to about 3 inches. In some embodiments, the predetermined with may be about 2 and ¼ inches to about 2 and ¾ inches. In preferred embodiments according to the present invention the predetermined width is about 2 and ½ inches.

(8) It is within the scope of the present invention that the frame be not only used in windows and doors, but also in a variety sliding glass door assemblies having several different track configurations. The frame assembly according to the present invention may comprise a sill assembly and a header assembly both interconnected to a securing surface such as a floor, a wall, or a ceiling. The support assembly may be interconnected to a header assembly. The header assembly supports and guides the frame, and more specifically the support assembly. More than one header assembly may also be provided according to the present invention. A header assembly may be attached to a ceiling, while a different header assembly may be attached to a wall. A portion of the frame, and more specifically the support assembly, comprises a track assembly. The sill assembly generally comprises a sill having a roller member and a roller track. Together, the roller member and the track assembly are cooperatively configured so that the roller member supports the track assembly in a movement permitting relation. The frame of the present invention may be moved relative to the header assembly and the sill assembly. The frame assembly may include more than

one frame movable relative to the header assembly and the sill assembly. Movement of the frame relative to the header assembly and the sill assembly may be facilitated by a frame handle which may be movably or removably attached to the frame.
(9) It is also within the scope of this invention to provide a track and a header assembly that can accommodate more than one frame via a header assembly and a corresponding sill assembly having multiple tracks each configured to movably retain one frame. The header assembly and corresponding sill assembly may have a configuration having either two, three, or four tracks. More than four tracks may also be possible depending on the need. Some track configurations may comprise a plurality of frames disposed in adjacent relation to one another within the same track. In some embodiments according to the present invention the frame assembly may comprise an interconnecting member that interconnects two adjacent frames, or that interconnects a frame to a securing surface.
(10) These and other objects, features and advantages of the present invention will become clearer when the drawings as well as the detailed description are taken into consideration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For a fuller understanding of the nature of the present invention, reference should be had to the following detailed description taken in connection with the accompanying drawings in which:
(2) FIG. **1** is a section view of one preferred embodiment of the frame of the frame assembly according to the present invention.
(3) FIG. **2** is a section view of another preferred embodiment of the frame of the frame assembly according to the present invention.
(4) FIG. **3** is a section view of a different preferred embodiment of the frame of the frame assembly according to the present invention comprising a strengthening member.
(5) FIG. **4** is a perspective exploded view in partial cutaway of one preferred embodiment of a part of a window frame of the present invention comprising two frames disposed adjacent to one another and a strengthening member.
(6) FIG. **5**A is a perspective exploded view of one preferred embodiment of part of a window frame of the present invention.
(7) FIG. **5**B is an assembled perspective view of the embodiment as shown in FIG. **5**A.
(8) FIG. **6**A is a section view of one preferred embodiment of the frame assembly of the present invention comprising a frame disposed on a header assembly.
(9) FIG. **6**B is a section view of another preferred embodiment of the frame assembly of the present invention comprising a frame disposed on a header assembly and a lock assembly.
(10) FIG. **7** is a section view of one preferred embodiment of the frame assembly of the present invention comprising a frame disposed on a sill assembly.
(11) FIG. **8**A is an exploded perspective view of one preferred embodiment of a frame of a sliding glass door of the present invention.
(12) FIG. **8**B is an assembled perspective view of the embodiment as shown in FIG. **8**A.
(13) FIG. **8**C is an exploded perspective view of one preferred embodiment of a header assembly and a sill assembly of a sliding glass door of the present invention.
(14) FIG. **8**D is an assembled perspective view of the embodiment as shown in FIG. **8**C.
(15) FIG. **9** is a section view of one preferred embodiment of part of a sliding glass door according to the present invention comprising three tracks.
(16) FIG. **10**A is a top section view of one preferred embodiment of a sliding glass door according to the present invention comprising two tracks and two frames.
(17) FIG. **10**B is a top section view of another preferred embodiment of a sliding glass door

according to the present invention comprising two tracks and three frames.
(18) FIG. **10**C is a top section view of a different preferred embodiment of a sliding glass door according to the present invention comprising two tracks and four frames.
(19) FIG. **10**D is a top section view of another preferred embodiment of a sliding glass door according to the present invention comprising two tracks and four frames.
(20) FIG. **10**E is a top section view of another preferred embodiment of a sliding glass door according to the present invention comprising two tracks and two frames.
(21) FIG. **10**F is a top section view of a preferred embodiment of a sliding glass door according to the present invention comprising one track and one frame.
(22) FIG. **10**G is a top section view of a preferred embodiment of a sliding glass door according to the present invention comprising two tracks and six frames.
(23) FIG. **10**H is a top section view of another preferred embodiment of a sliding glass door according to the present invention comprising three tracks and three frames.
(24) FIG. **10**I is a top section view of yet another preferred embodiment of a sliding glass door according to the present invention comprising three tracks and three frames.
(25) FIG. **10**J is a top section view of a different preferred embodiment of a sliding glass door according to the present invention comprising three tracks six frames.
(26) FIG. **10**K is a top section view of a preferred embodiment of a sliding glass door according to the present invention comprising four tracks and eight frames.
(27) FIG. **10**L is a top section view of another preferred embodiment of a sliding glass door according to the present invention comprising four tracks and four frames.
(28) FIG. **10**M is a top section view of a yet another preferred embodiment of a sliding glass door according to the present invention comprising four tracks and four frames.
(29) FIG. **10**N is a top section view of a different preferred embodiment of a sliding glass door according to the present invention comprising four tracks and eight frames.
(30) FIG. **11**A is a top section view of a one preferred embodiment of a sliding glass door according to the present invention comprising a strengthening member.
(31) FIG. **11**B is a top section view of a one preferred embodiment of a sliding glass door according to the present invention comprising a strengthening member and a plurality of frame handles.
(32) FIG. **12** is a top section view of one preferred embodiment of a sliding glass door according to the present invention comprising two interconnected frames.
(33) FIG. **13**A is a top section view of one preferred embodiment of a sliding glass door according to the present invention comprising an interconnecting member disposed on a support assembly.
(34) FIG. **13**B is a top section view of one preferred embodiment of a sliding glass door according to the present invention comprising an interconnecting member disposed on a strengthening member.
(35) FIG. **14**A is a section view of one preferred embodiment of the sill assembly and portions of the track assembly in one orientation.
(36) FIG. **14**B is a section view of one preferred embodiment of the sill assembly and portions of the track assembly in another orientation.
(37) Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(38) The present invention is directed to a frame assembly **1** for use in a variety of applications including doors and windows for buildings. The frame assembly **1** of the present invention may also be used in connection with other applications and is not limited to buildings. Generally, the frame assembly **1** comprises a frame **10** structured to retain a panel **12**. The panel **12** may be made of a material such as, but not limited to, glass. The frame assembly **1** according to the present invention comprise a frame **10** that can retain a panel **12**. The panel **12** may have impact resistant properties.

(39) As represented in FIG. **1**, at least a portion of the frame **10** has a predetermined thickness **14** and a predetermined width **16**. The predetermined thickness **14** is substantially less than the predetermined width **16**. As a result, the frame **10** has a predetermined thickness **14** with a reduced dimension that is substantially less relative to a purposefully increased dimension of the predetermined width **16**. Viewing through the panel **12**, such as along a normally intended line of sight, schematically indicated as **200** in one direction and as **200′** in another direction, is enhanced by concurrently and cooperatively maximizing the visible area of the panel **12** relative to the visible area of the frame **10**. Viewing through the panel **12**, and in more specific terms viewing along a normally intended line of sight **200** or **200′**, is enhanced by reducing the size of the predetermined thickness **14**, due to the fact that a lesser or smaller part of a visually exposed portion of the predetermined thickness **14**, as at **13**, will be disposed in overlying, visually obstructing relation to the panel **12**. It is further recognized that one possibility and/or consequence of reducing the size of the predetermined thickness **14** is that this may derogatorily affect the overall stability of the frame **10** and or at least a portion of the frame **10** which includes the reduced size and/or configuration of the predetermined thickness **14**, as indicated.

(40) Therefore, in order to enhance the stability of the frame **10** or at least maintain an appropriate stability thereof, the predetermined width **16** of the same or corresponding portion **13′** of the frame **10** is substantially increased. As should be apparent from a view of at least FIG. **1**, an increase in the predetermined width **16** of the same and/or corresponding portion **13′** will not interfere with the viewing of or through the panel **12**, since the increased width **16** is substantially or at least partially aligned with the aforementioned line of sight **200** or **200′**.

(41) Therefore, the structural integrity of the frame **10** is also enhanced, or at least maintained, since the relatively increased width **16** of the frame **10** compensates for its relatively decreased thickness **14** and provides an adequate, appropriate and/or required structural stability. Generally, at least a portion **13** of the side of the frame **10**, and a portion **13′** of the base of the frame **10**, has the cooperatively dimensioned, decreased predetermined thickness **14** and correspondingly increased predetermined width **16**, respectively. However, the majority, or even all, of the portion **13** and **13′** of the frame **10** may have the cooperatively dimensioned, decreased predetermined thickness **14** and correspondingly increased predetermined width **16**, respectively.

(42) Turning in detail to the frame assembly **1** of the present invention, as at least represented in FIG. **1**, the frame **10** generally comprises a cover assembly **20** interconnected to a support assembly **30**. But, the frame could instead be a single piece. The cover assembly **20** and the support assembly **30** may be made out of a variety of materials including, but not limited to, aluminum. The cover assembly **20** generally comprises a cover structure **22**. The cover structure **22** may be in the form of a plate, as represented in FIGS. **1-3**, or may be in the form of a glass bead, as represented in **6**A, **6**B, **7**, **9**, **11**A, **11**B, **12** and **13**. The cover structure **22**, whether in the form of a plate or a glass bead, may be made out of a variety of materials such as, but not limited to, aluminum. The cover structure **22** may comprise a plurality of latching members, generally indicated as **24**. The cover assembly **20** may also comprise a retaining member **26**. Attached to the retaining member **26** may be a cover clamp, generally indicated as **28**. The support assembly **30** may have a plurality of latching structures, generally indicated as **34**, which are correspondingly positioned to the plurality of latching members **24** of the cover assembly **20**. The plurality of latching members **24**, and correspondingly disposed latching structures **34**, form a mating engagement to interconnect the cover assembly **20** to the support assembly **30**. The support assembly **30** is generally connected to a securing surface **18**. This securing surface **18** may be a wall, a ceiling, a flooring surface, a soffit, or a different type of surface that supports the frame **10**. A variety of connectors, indicated as **19**, in FIGS. **13**A and **13**B, such as concrete or drywall screws or fasteners may be used to interconnect the support assembly **30** to the securing surface **18**. Preferred embodiments according to the present invention may include 1 and ½ tapcons to attach the frame **10** to the securing surface **18**, but other fasteners may also be used depending on the substrate.

(43) Generally, the cover assembly **20** and the support assembly **30** are cooperatively configured to retain panels **12** which may have varying dimensions. More specifically, the retaining member of the cover assembly **20**, and a cap segment **36** of the support assembly **30**, are disposed in spaced relation to receive panels **12** of varying dimensions therebetween. In some embodiments according to the present invention, the cover clamp **28** may be disposed in spaced relation relative to the cap segment **36** of the cover assembly **20**. The cover clamp **28** may be structured to receive a sealant member **15** that provides a seal to the panel **12**. The sealant member **15** may be in the form of caulking, silicone, or other sealant. In preferred embodiments where the frame **10** is made of a hollow material, stability of the frame **10** may be further enhanced by providing a support element **35**. By way of example, the support element **35** as shown in FIG. **1** may be a wood buck disposed inside of the frame **10**, and more specifically inside the support assembly **20**.

(44) As represented in at least FIG. **2**, the frame assembly **1** according to the present invention may also include a base **40** structured to support the frame **10**. The base **40** may be attached to the securing surface **18** via connectors or via a resin-based material such as, but not limited to, epoxy. The base **40** may have a plurality of protrusions **48** and a recessed portion **42**. The frame **10**, and in some embodiments the support assembly **30**, may have a plurality of grooves **38** that correspond to the protrusions **48** of the base **40**. The grooves **38** generally has an opening of sufficient size to receive the protrusions **48**. A sealant such as, but not limited to, silicone may be used to secure the plurality of protrusions **48** to the corresponding groove **38**. A sealant such as, but not limited to, silicone may also be applied within the recessed portion **42** to secure the frame **10** to the base **40**. With further reference to the illustrative example as represented in FIGS. **1** and **2**, which show a frame **10** having a cover assembly **20** interconnected to a support assembly **30** attached to a base **40**, the predetermined thickness **14** of the frame **10** is substantially less than the predetermined width **16**. In the illustrative embodiment as shown in FIG. **2**, the base **40** is supported by a support element **45** in the form of a wood buck.

(45) In at least one preferred embodiment, the predetermined thickness **14** is no greater than substantially 1 and ½ inches. In other embodiments, the predetermined thickness **14** may be generally about 1 to 1 and ½ inches. In some preferred embodiments, the predetermined thickness **14** may be about 1 and 3/16 inches. The predetermined width **16** may be generally about 2 to about 3 inches. In some embodiments, the predetermined width **16** may generally be about 2 and ¼ inches to **2** and ¾ inches. Other preferred embodiments may comprise a frame **10** with a predetermined width **16** of about 2 and ½ inches. Several other combinations between the predetermined thickness **14** and predetermined width **16** are also possible.

(46) The frame assembly **1** according to the present invention may comprise a plurality of frames **10** disposed in adjacent relation to one another as is shown in the illustrative embodiment of FIG. **3**. Preferred embodiments according to the present invention may include two frames **10** interconnected by a strengthening member **50**. The strengthening member **50** may comprise an interconnecting portion **52** and a supporting portion **54**. The interconnected portion **52** may comprise a serrated or uneven surface to facilitate application of a sealant or other binding material. As such at least a portion of the support assembly **30** of each frame **10** interconnects to the interconnection portion **52** of the strengthening member **50**. Other embodiments according to the present invention may comprise a plurality of frames **10** directly interconnected to each other without a strengthening member **50**. The strengthening member **50** generally has a supporting portion **54** which may comprise a plurality of cells **56**. The cells **56** may be reinforced or unreinforced depending on the need. Additionally, the strengthening member **50** may be interconnected to a support plate **58**. The support plate **58** may be connected to the securing surface **18** by connectors or by a resin-based material. Additionally, the support plate **58** may have a plurality of flanges **59** each of which corresponds to one of the plurality of cells **56** of the strengthening member **50**. As such, each of the plurality of cells **56** may be have a dimension sufficient to receive a corresponding one of the plurality of flanges **59** of the support plate **58**.

(47) To further improve the aesthetic appearance of the frame assembly **1**, a cover cap **57** may be disposed one side of the frame **10**. The illustrative embodiment of FIG. **3** shows two frames **10** interconnected by a strengthening member **50**, and a cover cap **57** disposed on one end of the frames **10**. The illustrative embodiment as shown in FIG. **4** shows a portion of two frames **10** interconnected by a strengthening member **50** and a support plate **58** disposed on one end of the strengthening member **50**. As can be appreciated, the support plate **58** and the strengthening member **50** cooperatively enhance the stability of the frame(s) **10** by adding further points of attachment to the securing surface **18** and by providing an additional structural element along the length of the frame **10**. Additional structural members may be added to the frame **10**, and in some embodiments, specifically to the support assembly **30**. The Illustrative embodiment as represented in FIG. **5**A shows an angle **53** disposed within adjoining sections of the frame **10**. Further, the illustrative embodiment as represented in FIG. **5**A also shows two adjoining sections of a frame **10** interconnected by a coupler **55**.
(48) As represented in at least FIG. **9**, the frame assembly **1** of the present invention may also be configured for use as a sliding glass door. As represented in FIGS. **6-9**, the present invention may comprise a header assembly and a sill assembly, generally indicated as **60** and **70** respectively. The header assembly comprise a header member **62**, a header base **61**, a header cover and a header support **68**. A plurality of header connectors **64** may be disposed on one end of the support assembly **30** of the frame **10**. Furthermore, the header connectors **64** may comprise a retaining portion, generally indicated as **67**. As is shown in the illustrative embodiments of FIGS. **6**A and **6**B, the header connectors **64** may be disposed in spaced relation relative to each other and surrounding a header member **62**. To further facilitate alignment of the frame **10** relative to the header assembly, a plurality of corresponding alignment members **63** may be disposed on each of the header connectors **64**. More specifically the alignment members **63** may be disposed within the retaining portion **67**. The alignment members **63** may be in the form of a weather-stripping member. As can be appreciated in the illustrative embodiments represented in FIGS. **6**A and **6**B, a header support may be placed on the sill base **61** to reduce any gaps that may exist between the base **61** and the header connectors **64**. The header support **68** also permits movement of the frame **10** relative to the header base **61**. An additional sealant member **15′** may also be disposed on a different part of the frame **10**. Also, as represented in FIGS. **7-9**, the frame assembly **1** of the present invention may also comprise a sill assembly **70** which supports the frame **10**. Preferred embodiments according to the present invention comprise a frame assembly **1** having both a header assembly **60** to support a frame **10** on one end, and a sill assembly **70** to support a different frame **10** on another end. The header assembly **60** and the sill assembly **70** may both be attached to a securing surface **18**. By way of example only, the header assembly **60** may be attached to a ceiling, a wall, or a soffit, while the sill assembly **70** may be attached to a flooring surface such as a concrete slab. Generally, the frame assembly **1** comprises at least one frame **10** supported by both a header assembly **60** and a sill assembly **70**.
(49) The sill assembly **70** of the frame assembly **10** according to the present invention generally comprises a sill base **71** attached to a securing surface **18**, and a sill cover **74** having a plurality of tracks **84** formed thereon. The sill cover **74** may also have at least one expansion chamber formed therein. Alternatively, the tracks **84** formed on the sill cover **74** may at least partially comprise or define at least one expansion chamber **87**. A plurality of sill members **72** may also be formed on the sill base **71** to support a corresponding track **84**. The corresponding track **84** again, also able to at least partially comprise or define at least one expansion chamber **87**. The illustrative embodiment as represented in FIG. **7** shows a frame **10** disposed within a sill assembly **70** having two tracks **84** and two expansion chambers **87**. As can be appreciated from the illustrative embodiment as represented in FIG. **7**, a sill assembly **70** comprises a sill base **71** and a sill cover **74** interconnected to the sill base **71**. As can also be seen from the illustrative embodiment of FIG. **7**, the sill cover **74** comprises two tracks **84** formed thereon. Further, the two tracks preferably include expansion

chambers formed therein. Each track **84** is generally structured to receive a roller member **82**. The roller member **82** is rotationally connected to the sill assembly **70**. The roller members **82** are also at least partially retained by the expansion chambers **87**.

(50) With primary reference to FIG. **9**, the frame assembly **1** may comprise a track assembly substantially formed on a portion of the support assembly **30** of the frame **10**. The track assembly **80** may comprise a plurality of sill connectors **64′** disposed in spaced relation relative to one another. The sill connectors **64′** generally surround a portion of the track cover **74**. Similar to the header connectors **64** of the header assembly **60**, the sill connectors **64′** of the sill assembly **70** may each comprise retaining portion **67** and alignment members **63** disposed thereon. Additional elements such as a sill support **78** may also be disposed on the sill cover **74** to facilitate supporting the frame **10**. To improve the aesthetic appearance of the frame assembly **1**, the sill assembly **70**, and more specifically the sill base **76**, may comprise a sill segment **76** to cover at least a portion of the sill assembly **70**. Moreover, a sill extension **76′** may be disposed on the sill segment **76** further cover the sill assembly **70**.

(51) As also represented in FIGS. **7**, **9**, **14**A and **14**B the track assembly **80** generally comprises two tracks **84** and preferably two expansion chambers **87** at least partially interconnected to the sill assembly **70**. The tracks **84** and the expansion chambers **87** are disposed on the roller member **82**. Generally, the frame **10**, and more specifically the tracks **84** of the track assembly **80**, are movable relative to the sill assembly **70**. As such, in preferred embodiments according to the present invention, the frame **10** is movable relative to both the sill assembly **70** and the header assembly **60**. Some embodiments according to the present invention may comprise a plurality of frames **10** disposed within the header assembly **60** and the sill assembly **70**.

(52) With primary reference to FIGS. **14**A and **14**B, the sill assembly **70** may comprise at least one expansion chamber **87**, wherein the sill cover **74** at least partially defines an expansion chamber **87**. In an alternative embodiment, and by way of example, the tracks **84** formed on the sill cover **74** may comprise or define an expansion chamber **87**, the expansion chamber **87** structured to retain, receive, operatively connect or rotationally connect at least a portion of a roller member **82** or other portions of the track assembly **80**.

(53) By way of a non-limiting example, the roller member **82** when retained, received, operatively connected, or rotationally connected at least to a portion of the expansion chamber **87**, may exert a force upon the tracks **84** or any other portion of the sill assembly **70** that the roller member **82** is in contact with. Such a force or contact may induce frictional forces that hinder the ability of the track assembly **80** to move or operate properly in relation to the sill assembly **70**. In another embodiment, the force or contact may induce forces upon the sill assembly **70** or portions thereof.

(54) As such, expansion chambers **87** are included and structured at least to expand from a constricted or resting state **87′** to an expanded state **87″**. At least when the expansion chamber **87** is comprised or defined by the tracks **84**, the expansion chamber **87** can expand from a resting state **87′** to an expanded state **87″**. In such a situation, upon expansion of the expansion chamber **87**, the tracks **84** also expand from a resting state to an inflated state. The expansion of the tracks **84** is attributable to a displacement in volume of the expansion chamber **87**, at least when in said expanded state **87″**.

(55) In one embodiment, expansion of the expansion chamber **87** occurs at least partially when a force is inflicted on the expansion chamber **87** or surrounding areas of the sill cover **74** by a portion of the track assembly **80**. In the same or other embodiments, expansion of the expansion chamber **87** may reduce frictional forces existing between a portion of the sill cover **74** and the track assembly **80**. In the same or other embodiments, expansion of the expansion chamber **87** may more equally distribute forces inflicted upon the sill cover **74** via the track assembly **80**.

(56) It is within the scope of the present invention that the frame **10** of the present invention be used not only in windows but also in sliding glass door assemblies. It is also within the scope of the present invention to provide a frame assembly **1** comprising a plurality of frames **10** for use as

sliding glass doors. Both the header assembly **60** and the sill assembly **70** may be configured to support more than one frame **10**. As such, the sill assembly **70** may comprise a plurality of tracks **84** and/or a plurality of expansion chambers **87** while the header assembly **60** may comprise a plurality of corresponding header members **62**. The illustrative embodiment as shown in FIGS. **6**A and **6**B shows a header assembly **60** having two header members **62** while the illustrative embodiment as represented in FIG. **7** shows a sill assembly **70** having two tracks **84** and two expansion chambers **87** that correspond to each frame **10** and each header member **62**. Therefore, two frames **10** may be movably disposed on the header assembly **60** and the sill assembly **70**. It is also possible, and within the scope of the present invention, to provide a frame assembly **1** having a header assembly **60** and a sill assembly **70** configured to receive more than two frames **10**. As such the frame assembly **1** of the present invention may be used as a sliding glass door assembly having multiple tracks **84** and multiple expansion chambers **87** with corresponding frames **10**. In preferred embodiments according to the present invention, the frames **10** may be movable.
(57) Some embodiments according to the present invention may comprise more than one header assembly **60**. By way of example, FIGS. **8**A-**8**D shows a portion of a frame assembly **1** comprising two header assemblies **60** and a sill assembly **70**, with a corresponding frame **10**. The illustrative embodiment as shown in FIG. **9** shows a frame assembly **1** having three tracks **84**. As can be appreciated from the illustrative embodiment of FIG. **9**, each of the three tracks **84** with corresponding roller members **82** supports a frame **10**. Further embodiments comprising a combination between two to four or more tracks **84**, and one to eight or more frames **10**, are also possible. The illustrative embodiments as shown in FIGS. **10**A-ION represent some of the combinations between tracks **84** and frames **10** that can be achieved.
(58) Additional structural features of the present invention include a frame handle, generally indicated as **90** in FIG. **11**A. The frame handle **90** may include a handle segment **92**. The handle segment **92** may comprise a handle surface **94** interconnected to a portion of a support assembly **30**. The handle segment **92** may also comprise a handle connector **96** while the support assembly may have a corresponding socket configured for insertion of the handle connector **96**. In some embodiments according to the present invention, the handle **90** may be interconnected to the frame **10** by a screw or other type of connector as is the case with the illustrative embodiments as shown in FIGS. **12** and **13**. In other embodiments according to the present invention, the handle connector **96** may be removably interconnected to the socket of the support assembly **30**. With reference to FIGS. **11**A and **11**B, the frame assembly **1** comprises two frames **10** disposed in adjacent relation to one another wherein one frame **10** is interconnected to a strengthening member **50**, and wherein the other frame **10** is movable relative to the strengthening member **50**. In the illustrative embodiments of FIGS. **11**A and **11**B, the support assemblies **30** and the strengthening member **50** are interconnected through a lock assembly **98**. The lock assembly **98** may comprise a bolt or other locking mechanism which may be pushed to lock or release one or more frames **10** from the locking member **50**. The illustrative embodiment as shown in FIG. **11**A also shows two support assemblies each having a socket for insertion of the handle connector **96** of a handle **90**. The illustrative embodiment of FIG. **11**A also shows two handles each having a handle connector interconnected to the socket of its corresponding support assembly **30**.
(59) Yet additional features of the frame assembly **1** of the present invention include an interconnecting segment **104** formed on the frame **10**. As is shown in FIG. **12**, the interconnecting segment **104** may be formed on the support assembly **30** and interconnects two frames **10**. The illustrative embodiment of FIG. **12** shows two interconnected frames **10** each one having an interconnecting segment **104** disposed in mating engagement with one another thereby interconnecting the two frames **10**. The two frames **10** as shown in FIG. **12** are movable relative to one another. Additionally, the interconnecting segment **104** may be attached the securing surface **18** as is shown in FIG. **13**A. FIG. **13**A shows a frame **10** having an interconnecting segment **104** attached thereto by a connector **19**. In the illustrative embodiment as shown in FIG. **13**A, the frame

10 is movable relative to the securing surface **18**. Moreover, FIG. **13**B shows an illustrative embodiment wherein a frame **10** has an interconnecting segment **104** attached to a strengthening member **50** so that a user may be able to see a portion of the frame **10** when the assembly **1** is fully closed.

(60) Since many modifications, variations and changes in detail can be made to the described preferred embodiment of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Thus, the scope of the invention should be determined by the appended claims and their legal equivalents.

Claims

1. A frame assembly for retaining a panel comprising: a frame comprising a support assembly and a cover assembly interconnected to one another in retaining relation to the panel, said support assembly comprising a predetermined thickness and a predetermined width; said predetermined thickness being of a substantially lesser dimension than said predetermined width, said predetermined thickness of said support assembly disposed in viewable relation to an external, normal line of sight of the panel and being dimensioned to enhance viewing of a greater portion of the panel, at least along said external, normal line of sight, a track assembly comprising at least one track and at least one roller member, said at least one roller member movable relative to said at least one track, said predetermined thickness and said predetermined with of said support assembly being relatively disposed and cooperatively dimensioned to maintain stability of said frame, and at least one expansion chamber disposable between an expanded state and a constricted state concurrent to forces being exerted on said at least one track by said at least one roller member.
2. The frame assembly as recited in claim 1 wherein said cover assembly comprises a cover structure interconnected to said support assembly, said cover structure cooperatively structured with said support assembly to collectively retain the panel on said frame.
3. The frame assembly as recited in claim 2 further comprising a retaining member and a cap segment formed on said cover structure; said retaining member and said cap segment cooperatively disposed to collectively facilitate retention of the panel on said frame.
4. The frame assembly as recited in claim 1 wherein said predetermined thickness of said support assembly is no greater than 1½ inches.
5. The claim assembly as recited in claim 4 wherein said predetermined width of said support assembly is generally between 2 inches and 3 inches.
6. The claim assembly as recited in claim 1 wherein said predetermined with of said support assembly is generally between 2 inches and 3 inches.
7. The claim assembly as recited in claim 1 wherein said cover assembly and said support assembly are interconnected to one another on opposite sides of the panel in collectively retaining relation thereto.
8. The claim assembly as recited in claim 1 further comprising a plurality of sill connectors disposed in spaced relation to one another, each of said plurality of sill connectors structured to surround a portion of a sill cover.
9. The claim assembly as recited in claim 8 including a sill base structured to attach to a securing surface; said plurality of sill connectors formed on said sill base.
10. The claim assembly as recited in claim 9 wherein said plurality of sill connectors are structured to support said at least one track.
11. The claim assembly as recited in claim 1 further comprising a sill assembly including a plurality of sill connectors, said plurality of sill connectors including at least one retaining portion and at least one alignment member, said at least one alignment member disposed on said at least one retaining portion.

12. The claim assembly as recited in claim 1 wherein said at least one expansion member is disposed and structured to expand said at least one track from a resting state, upon a displacement in volume of said at least one expansion member, concurrent to said expanded state thereof.
13. A frame assembly for retaining a panel comprising: a frame comprising a support assembly and a cover assembly interconnected to one another in retaining relation to the panel, said support assembly comprising a predetermined thickness and a predetermined width; said predetermined thickness being of a substantially lesser dimension then said predetermined width, said predetermined thickness of said support assembly disposed in viewable relation to an external, normal line of sight of the panel and being dimensioned to enhance viewing of a greater portion of the panel, a sill assembly structured to support said frame and including a sill cover, said sill cover including at least one track and at least one roller member, said predetermined thickness and said predetermined with of said support assembly being relatively disposed and cooperatively dimensioned to maintain stability of said frame, and said at least one track including at least one expansion member disposable between an expanded state and a constricted state concurrent to forces being exerted on said at least one track by said at least one roller member.
14. The frame assembly as recited in claim 13 wherein said cover assembly comprises a cover structure interconnected to said support assembly, said cover structure cooperatively structured with said support assembly to collectively retain the panel on said frame.
15. The frame assembly as recited in claim 14 wherein said at least one expansion member is disposed and structured to expand said at least one track from a resting state upon a displacement in volume of said at least one expansion member, concurrent to said expanded state thereof.
16. The frame assembly as recited in claim 14 wherein said at least one expansion chamber is structured to retain at least a portion of said at least one roller member.
17. The frame assembly as recited in claim 14 wherein said expanded state of said at least one expansion member is structured to distribute forces upon said sill cover.
18. The frame assembly as recited in claim 14 wherein said sill assembly includes a sill base and a plurality of sill members; said plurality of sill members structured to support said at least one track.
19. The frame assembly as recited in claim 18 wherein said sill assembly further comprises a plurality of sill connectors including at least one retaining portion and at least one alignment member disposed on said at least one retaining portion.